Configuring SAML SSO for NetCore Meet Sites

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Summary

This article describes how to configure SAML 2.0 single sign-on for a NetCore Meet 5.0 site using NetCore Control Hub and a corporate identity provider. It covers service provider metadata export, IdP metadata import, attribute mapping, test sign-in, enforcement, and SAML signing certificate rollover, including troubleshooting for MTG-SSO-6001 and MTG-SSO-6004.

Applies to: NetCore Meet 5.0, NetCore Control Hub, NetCore Meet desktop app for Windows and macOS, NetCore Meet mobile app, NetCore Meet web client, Meet Starter, Meet Business, Meet Enterprise

Overview

NetCore Meet 5.0 supports SAML 2.0 SSO for user authentication through NetCore Control Hub. In this configuration, NetCore Meet acts as the SAML service provider, and the corporate identity platform acts as the identity provider.

A complete SSO configuration requires metadata exchange, attribute mapping, a successful test assertion, and explicit enforcement. Do not enforce SSO until at least one administrator has completed a successful SAML login test.

The most common SAML-related errors are MTG-SSO-6001 SAML Assertion Invalid or Expired and MTG-SSO-6004 IdP Certificate Mismatch. For the full error reference, see KB-MTG-2008.

- SAML binding: HTTP-Redirect for AuthnRequest and HTTP-POST for SAML Response.
- NameID format: email address.
- Required attributes: email, givenName, surname.
- Recommended optional attributes: displayName, department, ncMeetRole.
- Supported enforcement modes: Monitor, Enforced for selected groups, Enforced for all users.

Recommended SAML attribute mapping for NetCore Meet 5.0

NetCore Meet field	SAML attribute	Required	Example value
User principal	NameID	Yes	alex.chen@corp.example
Email	email	Yes	alex.chen@corp.example
First name	givenName	Yes	Alex
Last name	surname	Yes	Chen
Display name	displayName	No	Alex Chen
Department	department	No	Network Engineering
Meet role	ncMeetRole	No	meet.admin

Prerequisites

Complete the following prerequisites before configuring SAML SSO. The administrator performing the configuration must be able to access NetCore Control Hub and the corporate IdP administration console.

Maintain at least one break-glass local administrator account in NetCore Control Hub until SSO enforcement has been verified. Store the account credentials according to your organization’s privileged-access policy.

- NetCore Meet site is active on release 5.0.
- Administrator has NetCore Control Hub tenant-admin rights.
- Corporate IdP supports SAML 2.0 metadata import and signed SAML responses.

- Corporate IdP has an active signing certificate with a known expiration date.
- User directory contains valid email addresses for all NetCore Meet users.
- Client access to NetCore Control Hub over TCP 443 is permitted.
- System clocks on the IdP and NetCore Control Hub are synchronized. A clock skew greater than 5 minutes can cause MTG-SSO-6001 SAML Assertion Invalid or Expired.

```
# Optional pre-check from an admin workstation
ncc login --tenant acme --profile control-admin
ncc meet site show --site acme-prod

# Expected key fields
# Site ID        : acme-prod
# Release        : 5.0
# SSO Mode       : local
# Admin Role     : tenant-admin
# Control Hub    : reachable
```

Step-by-step Procedure

Use the following procedure to configure SAML SSO for a NetCore Meet 5.0 site. The examples use the site ID acme-prod and a corporate IdP entity ID of https://idp.corp.example/saml2.

The same configuration applies to the NetCore Meet desktop app, mobile app, and web client because authentication is brokered through NetCore Control Hub.

- 1. In NetCore Control Hub, go to Meet > Sites > acme-prod > Authentication > SAML SSO.
- 2. Set SSO mode to Monitor. Monitor mode allows SAML testing without blocking existing local sign-in.
- 3. Export the NetCore Meet service provider metadata. Save the XML file and import it into the corporate IdP as a new SAML application.
- 4. In the IdP SAML application, configure the NameID format as email address and set the NameID value to the user’s primary corporate email address.
- 5. Configure SAML response signing. NetCore Meet 5.0 requires the IdP to sign the SAML response or assertion.
- 6. Configure the required attributes: email, givenName, and surname. Add displayName, department, and ncMeetRole if your organization uses role mapping.
- 7. Export the IdP metadata XML from the corporate IdP.
- 8. Import the IdP metadata into NetCore Control Hub.
- 9. Save the attribute mapping and run a test sign-in with a non-break-glass administrator account.
- 10. After testing succeeds, change enforcement to either Enforced for selected groups or Enforced for all users.
- 11. Document the IdP signing certificate expiry date and configure a certificate rollover window at least 14 days before expiry.

```
# 1) Export NetCore Meet SP metadata
ncc meet sso metadata export \
  --site acme-prod \
  --format xml \
  --output ./netcore-meet-sp-acme-prod.xml

# Sample output
# Exported SP metadata for site acme-prod
# Entity ID : https://control.netcore.example/saml/sp/acme-prod
# ACS URL   : https://control.netcore.example/saml/acs/acme-prod
# SLO URL   : https://control.netcore.example/saml/slo/acme-prod
# File      : ./netcore-meet-sp-acme-prod.xml

# 2) Example IdP-side SAML application values to enter in the corporate IdP
cat ./netcore-meet-sp-acme-prod.xml | sed -n '1,18p'

# Expected metadata fields in the exported XML
# entityID="https://control.netcore.example/saml/sp/acme-prod"
```

```
# AssertionConsumerService Location="https://control.netcore.example/saml/acs/acme-prod"
# SingleLogoutService Location="https://control.netcore.example/saml/slo/acme-prod"

# 3) Import IdP metadata into NetCore Control Hub
ncc meet sso idp import \
  --site acme-prod \
  --metadata ./corp-idp-metadata-2026-02-01.xml

# Sample output
# IdP metadata imported
# IdP Entity ID       : https://idp.corp.example/saml2
# SSO URL             : https://idp.corp.example/sso/saml
# Signing Cert SHA256 : 8A:3D:91:2E:4C:73:6A:9F:55:42:BC:17:2D:90:01:AB:6C:11:ED:80:19:3B:EE:25:75:EF:D1:9A:
44:0C:62:10
# Certificate Expires : 2027-02-01T23:59:59Z

# 4) Configure attribute mapping
ncc meet sso attributes set \
  --site acme-prod \
  --nameid emailAddress \
  --map email=email \
  --map firstName=givenName \
  --map lastName=surname \
  --map displayName=displayName \
  --map department=department \
  --map role=ncMeetRole

# Sample output
# Attribute mapping saved for acme-prod
# Required mappings: complete

# 5) Keep SSO in Monitor mode for testing
ncc meet sso mode set --site acme-prod --mode monitor

# Sample output
# SSO mode for acme-prod set to monitor

# 6) Test SAML sign-in flow for a pilot user
ncc meet sso test \
  --site acme-prod \
  --user alex.chen@corp.example \
  --relay-state /meet/home

# Sample successful output
# Test result      : success
# NameID           : alex.chen@corp.example
# email            : alex.chen@corp.example
# givenName        : Alex
# surname          : Chen
# Assertion Issue  : 2026-02-03T14:21:08Z
# Assertion Expiry : 2026-02-03T14:26:08Z

# 7) Enforce SSO after successful validation
ncc meet sso mode set \
  --site acme-prod \
  --mode enforced-all \
  --confirm

# Sample output
# SSO mode for acme-prod set to enforced-all
# Break-glass local admin remains enabled: control-breakglass@corp.example

# 8) Configure planned certificate rollover
ncc meet sso cert-rollover prepare \
  --site acme-prod \
  --metadata ./corp-idp-metadata-2027-primary.xml \
  --activate-after 2027-01-18T02:00:00Z

# Sample output
# Rollover certificate staged
# Current Signing Cert SHA256 : 8A:3D:91:2E:4C:73:6A:9F:55:42:BC:17:2D:90:01:AB:6C:11:ED:80:19:3B:EE:25:75:EF:
D1:9A:44:0C:62:10
# Staged Signing Cert SHA256  : 6E:41:02:9A:10:BF:53:81:CC:8A:71:20:DE:4B:3F:98:01:E5:72:AA:49:17:C0:8B:7C:99:
2A:11:EF:63:5D:44
# Activation Time             : 2027-01-18T02:00:00Z
```

Verification

Verify SSO status from both the administrative plane and the user sign-in path. A successful configuration shows the correct IdP entity ID, valid signing certificate, complete attribute mapping, and enforced SSO mode.

If verification fails with MTG-SSO-6001 SAML Assertion Invalid or Expired, check assertion validity times, NameID, audience, recipient, and clock synchronization. If verification fails with MTG-SSO-6004 IdP Certificate Mismatch, re-import the current IdP metadata or activate the staged rollover certificate.

- Confirm SSO mode is enforced only after a successful pilot sign-in.
- Confirm the IdP certificate fingerprint in NetCore Control Hub matches the active IdP signing certificate.
- Confirm assertion audience equals the NetCore Meet SP entity ID.
- Confirm assertion recipient equals the ACS URL for the site.
- Confirm pilot users can sign in to the NetCore Meet desktop app, mobile app, and web client.

Verification checks and expected values

Check	Expected value
SSO mode	monitor during testing; enforced-selected-groups or enforced-all after cutover
SP Entity ID	https://control.netcore.example/saml/sp/acme-prod
ACS URL	https://control.netcore.example/saml/acs/acme-prod
IdP Entity ID	https://idp.corp.example/saml2
Required attributes	email, givenName, surname
Common failure code	MTG-SSO-6001 SAML Assertion Invalid or Expired
Certificate failure code	MTG-SSO-6004 IdP Certificate Mismatch

```
# Show current SSO configuration
ncc meet sso status --site acme-prod

# Sample output
# Site ID                 : acme-prod
# Release                 : 5.0
# SSO Mode                : enforced-all
# SP Entity ID            : https://control.netcore.example/saml/sp/acme-prod
# ACS URL                 : https://control.netcore.example/saml/acs/acme-prod
# IdP Entity ID           : https://idp.corp.example/saml2
# IdP SSO URL             : https://idp.corp.example/sso/saml
# Signing Cert SHA256     : 8A:3D:91:2E:4C:73:6A:9F:55:42:BC:17:2D:90:01:AB:6C:11:ED:80:19:3B:EE:25:75:EF:D1:
9A:44:0C:62:10
# Certificate Expires     : 2027-02-01T23:59:59Z
# Attribute Mapping       : complete
# Last Successful Login   : 2026-02-03T14:27:44Z alex.chen@corp.example

# Review recent SAML authentication events
ncc meet events list \
  --site acme-prod \
  --category sso \
  --since 2026-02-03T14:00:00Z \
  --limit 5

# Sample output
# TIME                  USER                    RESULT   CODE           DETAIL
# 2026-02-03T14:27:44Z  alex.chen@corp.example  success  -              SAML login completed
# 2026-02-03T14:24:02Z  alex.chen@corp.example  failed   MTG-SSO-6001  SAML Assertion Invalid or Expired
# 2026-02-03T14:18:55Z  maya.rao@corp.example   failed   MTG-SSO-6004  IdP Certificate Mismatch

# Decode the latest failed SAML diagnostic bundle
ncc meet sso diagnostics collect \
  --site acme-prod \
  --user alex.chen@corp.example \
  --output ./sso-diag-acme-prod-20260203.zip

# Sample output
```

```
# Diagnostic bundle created: ./sso-diag-acme-prod-20260203.zip
# Included: authn-request.xml, saml-response-redacted.xml, cert-fingerprint.txt, validation-report.txt
```

Rollback

If users cannot sign in after SSO enforcement, roll back to Monitor mode or local authentication. Use Monitor mode first when you want SAML events to continue logging while allowing local administrator access.

Do not delete the IdP configuration during an incident unless instructed by NetCore support. Preserving the configuration allows event correlation for MTG-SSO-6001 SAML Assertion Invalid or Expired and MTG-SSO-6004 IdP Certificate Mismatch.

- 1. Sign in with the break-glass local administrator account.
- 2. Change SSO mode from enforced-all or enforced-selected-groups to monitor.
- 3. Confirm administrators and pilot users can access NetCore Meet.
- 4. Review SSO events and correct the IdP metadata, certificate, or attribute mapping.
- 5. Re-test SAML sign-in before re-enforcing SSO.

```
# Roll back enforcement to Monitor mode
ncc login --tenant acme --profile breakglass-admin
ncc meet sso mode set --site acme-prod --mode monitor --reason "Rollback after SSO cutover issue INC-48291"

# Sample output
# SSO mode for acme-prod set to monitor
# Local sign-in allowed       : true
# SAML event logging retained : true

# If required, restore local-only authentication
ncc meet sso mode set --site acme-prod --mode local --confirm --reason "Temporary local-only recovery INC-
48291"

# Sample output
# SSO mode for acme-prod set to local
# SAML sign-in disabled for users
# IdP configuration retained  : true

# Confirm rollback state
ncc meet sso status --site acme-prod

# Sample output
# Site ID          : acme-prod
# SSO Mode         : local
# IdP Config       : retained
# Last Mode Change : 2026-02-03T15:02:19Z by control-breakglass@corp.example
```

Known Limitations

The following limitations apply to SAML SSO for NetCore Meet 5.0. Plan the deployment and certificate rollover process around these constraints.

For additional interpretation of SSO event codes, including MTG-SSO-6001 SAML Assertion Invalid or Expired and MTG-SSO-6004 IdP Certificate Mismatch, see KB-MTG-2008.

- NetCore Meet 5.0 supports SAML 2.0 for authentication only. It does not use SAML assertions for meeting authorization decisions such as host controls, meeting locks, or lobby admission.
- NameID must resolve to a valid user email address. Persistent opaque identifiers are not supported for first-time user matching.
- Signed SAML responses or signed assertions are required. Unsigned assertions are rejected.
- If the IdP signing certificate changes without metadata import or staged rollover, users may receive MTG-SSO-6004 IdP Certificate Mismatch.
- Assertions outside their validity window, assertions with incorrect audience or recipient, and assertions affected by clock skew may generate MTG-SSO-6001 SAML Assertion Invalid or Expired.

- SSO enforcement affects interactive sign-in for the NetCore Meet desktop app, mobile app, and web client. Existing active sessions remain valid until their normal session expiry.
- External meeting participants are not automatically provisioned by SAML SSO. In NetCore Meet 5.0, external participants wait in the lobby by default.
- Certificate rollover should be staged before the IdP begins signing with the new certificate. A 14-day operational window is recommended.

Related Articles

KB-MTG-2008: NetCore Meet and Calling Error Code Reference